Impaired Visuospatial Transformation but Intact Sequence Processing in Parkinson Disease

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Objective: We examined whether visuospatial deficits in Parkinson disease (PD) can be explained by a domain-general, nonspatial impairment in the sequencing or serial chaining of mental operations.

Background: PD has been shown to be associated with impaired visuospatial processing, but the mechanisms of this impairment remain unclear.

Methods: Thirteen patients with PD and 20 age-matched, neurologically normal controls performed a visuospatial grid navigation task requiring sequential spatial transformations. The participants also performed a control task of serial number subtraction designed to assess their nonvisuospatial sequencing. The tasks were matched in structure and difficulty.

Results: The patients were impaired on the visuospatial task but not in serial number subtraction. This finding suggests that visuospatial processing impairments in PD do not derive from a general impairment affecting sequencing or serial chaining.

Conclusions: We argue that visuospatial deficits in PD result from impairments to spatial transformation routines involved in the computation of mappings between spatial locations. These routines are mediated by dopaminergic pathways linking the basal ganglia, prefrontal cortex, supplementary motor area, and parietal cortex.

Key Words: Parkinson disease, visuospatial impairment, sequence processing

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ANOVA = analysis of variance. **msec** = milliseconds. **PD** = Parkinson disease. **SD** = standard deviation.

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Parkinson disease (PD) is a neurodegenerative illness caused by dopamine depletion in the basal ganglia and the consequent deregulation of striatal motor circuits (Fahn, 2003; Hu et al, 2006). As a motor disorder, PD causes slowness of movement, rigidity, and tremor, but it also causes impairments in executive function and control (Barnes and Boubert, 2008), working memory (Gabrieli et al, 1996; Kemps et al, 2005; Lee et al, 2010), speed of cognitive processing (Sawamoto et al, 2002), spatial attention (Pollux and Robertson, 2001; Possin et al, 2008), and language (Altmann and Troche, 2011; Murray, 2008).

PD has also been linked with impaired visuospatial processing. Typically, this deficit is measured by performance on tasks of classic mental rotation involving shape equivalence or mirror-image judgments on rotated visual images (Amick et al, 2006; Boller et al, 1984; Duncombe et al, 1994; Kerai et al, 2012; Lee et al, 1998; Levin et al, 1991).

An important unresolved question concerns the cognitive substrates of visuospatial impairment in PD. Studies of neurologically normal populations have shown that complex tasks like mental rotation are supported by several distinct, sequentially linked processes, eg, the encoding of display elements, their maintenance in spatial working memory, the application of image alignment or spatial transformation routines, and response selection (Leek et al, 2007; Shepard and Cooper, 1982). Task performance is also influenced by other stimulus-related factors such as object geometry, familiarity, complexity, and image-plane versus depth rotation (Arguin and Leek, 2003; Harris et al, 2008; Kerai et al, 2012; Leek 1998a, 1998b; Leek and Johnston, 2006; Tarr and Pinker, 1990). Thus, poor performance on a mental rotation task by patients with PD could reflect impairment at 1 or several processing stages, as well as differential sensitivity to other stimulus- and task-related factors.

In a 2012 study, Kerai et al contrasted the performance of patients with PD and age-matched, neurologically normal controls on 2 visuospatial processing tasks. One was a classic mental rotation task in which the participants judged the shape equivalence of 2 image-plane rotated novel objects. The other was a recognition memory task in which the participants had to match a rotated shape to a previously memorized vertically oriented stimulus. The key finding was that the patient group showed a steeper regression slope than the controls between reaction time and stimulus angular disparity.

This pattern is consistent with impairment seen in patients with PD, specifically of spatial transformation processes, rather than other task components (eg, feature encoding and maintenance in short-term memory, response selection and execution) that do not vary with the orientation of the stimulus (Johnson, 1990).

Our 2009 study showed that such spatial transformation processes involve mental routines that compute and execute mappings between visual images or spatial locations (Leek and Johnston, 2009). Our findings were consistent with Sawamoto et al (2002), who had found that patients with PD were impaired in a visuospatial grid navigation task. In Sawamoto's computerized "mental grid" task, participants were shown a series of 3 × 3 grids, each with 1 square marked as a start location. After seeing a grid, they were shown a sequence of arrows and were asked to imagine where the marked square would land if it moved in the directions shown by the arrows (Figure 1, Task 1). In other words, they were to execute a set of mental spatial shifts within the grid in response to the arrow cues, and to determine the end location. Like mental rotation, a mental grid task requires a spatial transformation—calculating and executing a mapping of shifts in location.

In the current study, we investigated whether the apparent impairment to spatial transformation processes in PD genuinely reflects an underlying visuospatial deficit, or instead arises from impairment to other, nonspatial processes that support visuospatial task performance, namely, the sequencing or serial chaining of mental operations. Sequence processing plays an important role in motor learning, the planning of complex actions, and many other nonmotor tasks involving the computation and execution of serial order. Visuospatial tasks like mental rotation and grid navigation may be assumed to have a sequencing or serial order component (Leek and Johnston, 2009). Interestingly, sequence processing has been linked to parts of the dopaminergic networks implicated in PD, including the supplementary motor area (Chung et al, 2005; Nachev et al, 2008), and mental rotation has itself been associated with motor simulation and activation within the premotor cortex (Johnston et al, 2004; Richter et al, 2000; Vingerhoets et al, 2001; Windischberger et al, 2003; Wohlschläger and Wohlschläger, 1998).

We tested the hypothesis that impairment to visuospatial transformation in PD derives from a nonspecific deficit in sequencing or serial chaining. We predicted that if this hypothesis were correct, then patients with PD who show impaired spatial transformation on visuospatial tasks should also show impairment in other, nonspatial sequencing tasks. To test this prediction, we examined the performance of patients with early-stage nondementing PD and age-matched neurologically normal controls on 2 carefully matched tasks. Task 1 assessed spatial transformation via visuospatial grid navigation, and Task 2 assessed nonspatial sequencing via serial number subtraction.

The visuospatial grid navigation Task 1, which we adapted from Sawamoto et al (2002), also required the participants to compute a sequence of shifts in location around a mental grid in response to a series of directional arrow cues. The nonspatial sequencing Task 2 used similar displays but required participants to subtract single digits from a starting number presented on the grid. While both tasks involved the sequential application of a mental operation (either spatial location shifts or subtraction), only Task 1 required spatial transformation.

The results showed that our patients with PD were impaired in the visuospatial grid navigation task, but not in sequential number subtraction, demonstrating that a nonspecific deficit in sequencing cannot account for visuospatial impairment in PD.

METHODS

Participants

We recruited 13 patients with a diagnosis of PD from a local PD clinic. The group had 1 woman and 12 men, with a mean age of 65.8 years, standard deviation (SD) = 7.9, and range 53 to 83 years. They had a Hoehn and Yahr (1967) score of 1 to 3 (mild to moderate PD) and a minimum score of 26/30 on the Mini-Mental State Examination (Folstein et al, 1975). We did both baseline and experimental testing when the patients were in their medication on-phase, approximately 90 minutes after their last regular dose.

We recruited 20 control volunteers from a database of local community residents who had expressed a willingness to participate in psychological studies. These 9 women and 11 men had no history of neurologic illness. With a mean age of 65.1 years, SD = 4.9, and range 51 to 79 years, we age-matched the controls to the patients.

All of the participants had normal or corrected-to-normal visual acuity. All had normal color vision according to the standard Ishihara color vision test.

Bangor University and National Health Service Research Ethics and Governance Committees approved the test protocol. We obtained informed consent from all participants.

Neuropsychological Profiles

We gave 4 additional baseline evaluation measures to all participants, to obtain a neuropsychological profile for each person and to facilitate comparisons of our patient sample across the literature. We gave the Mini-Mental State Examination (Folstein et al, 1975) to assess cognitive function, the Wechsler Abbreviated Scale of Intelligence (Wechsler, 1999) to measure verbal and performance intelligence, and the Benton Visual Retention Test (Benton Sivan, 1992) to measure visual perception and visual memory. In the Benton Visual Retention Test, the examiner shows participants a series of 10 cards, each containing 3 patterns. After displaying a card for 10 seconds, the examiner hides it and then asks the participants to draw the 3 patterns on paper as accurately as possible from memory. We assessed the participants' visual memory to rule out deficits in visuospatial memory that could affect their performance on the experimental tasks.

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Before starting the experimental tasks, we also assessed the participants' spatial short-term memory by giving them a task that used the same displays as in the 2 experimental tasks. We describe this task below in the *Task to Assess Spatial Short-Term Memory* section.

Table 1 shows the participants' mean scores on the baseline evaluation tests: spatial short-term memory, Mini-Mental State Examination, Benton Visual Retention Test, and Wechsler Abbreviated Scale of Intelligence. As a whole, the patient group scored within 2 standard deviations of the control group on all 4 measures. However, 2 of the patients performed significantly below the controls on the Wechsler Abbreviated Scale of Intelligence: Patient 9 on the verbal subtest and Patient 10 on the verbal, performance, and full intelligence quotient (IQ). (Bivariate correlations of the participants' experimental and baseline task performance are reported in the *Results* section and in Table 3 below.)

Experimental Tasks

Task 1 assessed spatial transformation through visuospatial grid navigation, and Task 2 assessed serial chaining through sequential number subtraction. The 2 tasks were matched for stimulus sequence lengths and used the same stimulus duration and offset-onset delay intervals (Figure 1).

Author J.H.K. performed all of the testing at the School of Psychology, Bangor University. Testing for each participant took 2 sessions. During Session 1 all participants completed the baseline tests. During Session 2 they completed the 2 experimental tasks in pseudo-random order. Each session took approximately 1 hour, with 30 minutes per experimental task.

Apparatus

We programmed the experiments using E-Prime® (Psychology Software Tools, Pittsburgh, Pennsylvania) and presented them on a 17-inch video graphics array monitor running at a screen resolution of 1024 × 768 pixels. Participants viewed the screen from a chinrest-controlled distance of 60 cm. The experimenter recorded participants' spoken responses using a microphone connected to a serial response box (Psychology Software Tools) that allowed for the measurement of voice-onset time. We defined voice-onset time as the interval between the appearance of the "response grid" (see next section) onscreen and the start of the participant's vocal response to seeing that grid.

Task 1: Visuospatial Grid Navigation

Each trial comprised a variable number of computer displays (Figure 1). The first ("cue grid") screen and final ("response grid") screen each showed a 3 × 3 square gray grid. Each grid subtended a visual angle of 17.62 degrees horizontally and 19.30 degrees vertically. In each trial, 1 of the 9 squares in the grid was highlighted; in each trial, it could be a different square.

In between the cue and response grids in each trial appeared a variable number of displays. In the center of some of the displays we showed a black arrow, which could be pointing left, right, up, or down. The left and right arrows

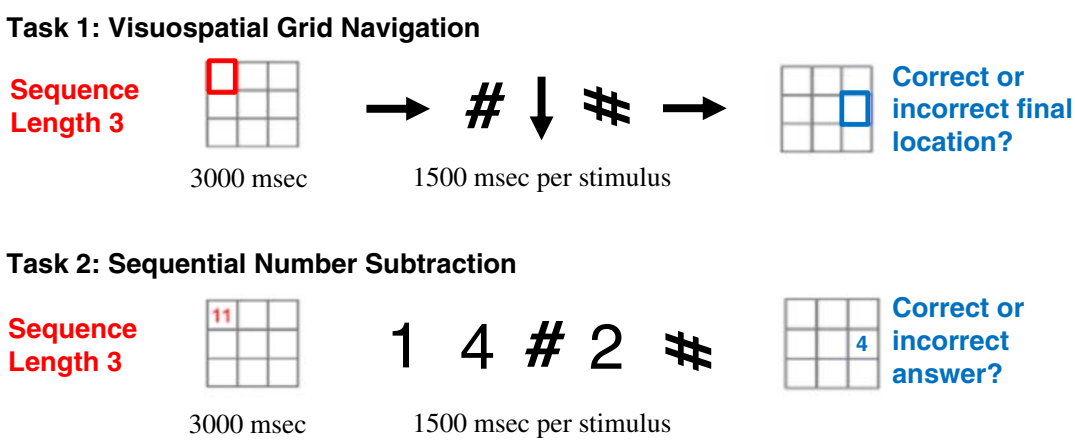

FIGURE 1. Schematic representation of sample trial sequences for the experimental tasks. Unlike this schematic, in the middle section of the actual experiment each arrow or digit and each placeholder appeared as a separate display.

Task 1: Visuospatial grid navigation. In the first of 3 sets of displays, participants saw a red-highlighted square within a grid. In the second series of displays, they saw a sequence of arrows, 1 at a time, and were told to calculate where the originally highlighted square would move if it followed the arrow sequence shown. (The participants were told to ignore the "placeholder" # symbol that was sometimes displayed instead of an arrow.) The final display showed a blue-highlighted square in the grid. The participants had to say whether or not the new highlight was in the location where their mental calculations had taken the original square.

Task 2: Sequential number subtraction. In the first of 3 sets of displays, participants saw a red number placed randomly within a grid. The second set of displays showed a series of digits, 1 at a time. Participants were told to subtract the sequence of digits from the starting number. (They were told to ignore the placeholder # symbol that was sometimes shown instead of a number.) The final display showed a blue number in the grid. Participants had to say whether this new number was the correct result of their serial subtractions.

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subtended 11.40 degrees horizontally and 8.58 degrees vertically. The up and down arrows subtended 8.58 degrees horizontally and 11.42 degrees vertically.

Sometimes instead of an arrow, in the center of a display we showed a black # symbol ("hash mark" or number sign), which subtended 10.85 degrees horizontally and 8.86 degrees vertically. The # symbols served as placeholders that let us vary the delay between displays of arrows, and to ensure that we showed the same total number of stimuli (arrows plus placeholders) in each trial. We showed the # symbol in 4 different forms: upright, rotated 90 degrees clockwise, and then both versions mirror-reversed. We altered the orientation to ensure that when we presented 2 placeholders immediately after 1 another, participants could distinguish between them.

For each trial, participants first saw the 3 × 3 square cue grid, with 1 corner square highlighted in red to denote the start location of the task. The experimenter told the participants to remember that location. After 3 seconds, the grid disappeared from the screen.

Following a blank inter-stimulus interval of 1500 milliseconds (msec), a variable sequence of 2, 3, or 4 arrows appeared, 1 at a time, each for 1500 msec. As noted, we pseudo-randomly interspersed 1 or more displays showing a placeholder. The experimenter told the participants to ignore the placeholders. She instructed the participants to recall the location of the highlighted red square that they had seen on the cue grid screen at the start of the trial, and then to follow the sequence of arrows in mentally moving the highlight from 1 square to another. When the sequence of arrows ended, the participants were to remember the last square that they had mentally highlighted.

After the last arrow or placeholder was shown, the screen went blank for another 1500 msec. Then the response grid screen appeared, with 1 square shown highlighted in blue. This grid remained onscreen until the serial response box recorded the participant's spoken "correct" or "incorrect" to the question of whether the highlighted blue square showed the location where the original red square would have landed after following the series of moves indicated by the arrows. The experimenter had instructed the participants to give their responses as quickly and accurately as possible.

This task had 3 trial length conditions: sequences of 2, 3, or 4 arrows, along with the pseudo-randomly interspersed placeholders, for 16 trials per condition. The start location was always 1 of the 4 corners of the grid; we tested each start location 12 times. Participants completed a total of 48 trials, presented in 2 blocks of 24 with a 2-minute rest period between them. We set half of the trials to require a "correct" response and the other half to require an "incorrect" response. Trial order was randomized within blocks, and the participants controlled the start time of each trial.

Task 2: Sequential Number Subtraction

Task 2 also comprised a variable number of displays. We used the same 3 × 3 grids as in Task 1, though for a different purpose. In the first display of this task, we

TABLE 1. Neuropsychological Test Scores for the Control and Parkinson Disease (PD) Groups

	Group Scores		Patients' Individual Scores												
	Control Mean (SD)	PD Mean (SD)	1	2	3	4	5	6	7	8	9	10	11	12	13
Sex			Man	Woman	Man	Man	Man	Man	Man	Man	Man	Man	Man	Man	Man
Hoehn and Yahr stage[1]		1.85 (0.73)	2	1	2	2	2	1	3	1	2	3	1	1	2
Subtype of PD			M	A/R	T	M	A/R	A/R	M	M	T	A/R	A/R	M	M
Disease duration (years)		9.85 (4.16)	10	7	8	8	11	8	21	7	7	15	12	7	7
Age (years)	65.1 (4.9)	65.84 (7.98)	70	72	67	59	64	74	64	58	64	65	83	53	63
Spatial short-term memory (% of 54 trials correct)	91.74 (4.04)	92.63 (1.91)	91.67	94.44	93.06	91.67	93.06	93.06	94.44	93.06	94.44	94.44	93.06	88.89	88.89
Mini-Mental State Examination[2]	29.8 (0.41)	29.15 (0.83)	28	29	28	29	29	28	30	30	30	30	29	30	29
Benton Visual Retention Test[3]	7.6 (1.59)	6.77 (1.76)	6	7	4	8	6	10	5	6	9	7	5	8	7
Wechsler Abbreviated Scale of Intelligence[4]															
Verbal IQ	125.46 (11.48)	113.31 (14.64)	109	119	104	136	128	107	128	114	89	91	119	102	127
Performance IQ	117.93 (17.37)	110.38 (25.80)	102	109	99	134	101	103	123	120	116	37	127	133	131
Full 4 IQ*	124.00 (12.88)	112.62 (17.56)	107	116	101	140	115	106	128	119	102	70	126	118	116

*This score combines the scores for verbal comprehension, perceptual reasoning, working memory, and processing speed.
[1]Hoehn and Yahr, 1967. [2]Folstein et al, 1975. [3]Benton Sivan, 1992. [4]Wechsler, 1999.
SD = standard deviation. **M** = mixed. **A/R** = akinetic/rigid. **T** = tremor-dominant. **IQ** = intelligence quotient.

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showed a cue grid with 1 randomly assigned square containing a red number from 9 to 18. Then we presented a variable sequence of digits, again 1 at a time. We used the digits 1 to 9, in black 48-point type, subtending about 1 degree horizontally and vertically. As in Task 1, we pseudo-randomly interspersed # placeholders among the digits. The placeholders were again black and could be upright, rotated, or mirror-reversed. The final display again showed a response grid, with 1 randomly assigned square containing a blue number from 1 to 16.

The first display, the cue grid with the red number, appeared onscreen for 3 seconds. The experimenter told the participants to memorize that number. Following a 1500-msec blank inter-stimulus interval, they saw the displays for the sequences of digits and placeholders, with each item appearing for 1500 msec. The participants were again told to ignore the placeholders. Their task was to recall the starting number and mentally to subtract from it all the digits that they saw in the sequence. When they saw the final display—the response grid containing the blue number—they were to say whether this number was the correct or an incorrect result of their sequence of subtractions. The response grid remained onscreen until their vocal response was recorded.

Task 2 had 3 trial length conditions: 2, 3, and 4 subtractions. Sequence length 2 had 2 serial subtractions from the original number. Sequence length 3 had 3 subtractions. Sequence length 4 had 4 subtractions. Participants completed a total of 48 trials, presented in 2 blocks of 24 with a 2-minute rest period between them. We gave 16 trials for each sequence length. Trial order was randomized within blocks, and the participants controlled the start time of each trial.

Task to Assess Spatial Short-Term Memory

This was the extra baseline task that we gave before the 2 experimental tasks, to assess the participants' spatial short-term memory. Using the same grids as in Tasks 1 and 2, we showed the participants a display with a "cue grid" with 1 of the 9 squares highlighted in red. We showed the stimulus for 1500 msec, followed by a blank 5-second retention interval. Then we presented a "response grid," again with 1 of the 9 squares highlighted in blue, either the same square as in the cue grid or a different square. The participants were instructed to say whether the second square was the same as or different from the first square.

This task had 54 trials, 6 for each square in the grid. This task told us whether the participants were able to retain memory for the location of a square over intervals that exceeded the maximum duration between cued events in the 2 experimental tasks.

Statistical Analysis

Our primary dependent measure was sensitivity (d'). We compared patient and control group discrimination, taking response bias into account. For each participant, we calculated the number of hits (H) and false alarms (FA) per condition. The difference between the z-transforms of these 2 rates determines the d' value:

$$d' = z(H) - z(\mathrm{FA})$$

(Macmillan and Creelman, 2005). We also analyzed mean voice-onset times. For all of the tasks, we adopted an a priori alpha level of 0.05. We report exact P values except where $P < 0.001$. We use $\eta^2_{\mathrm{partial}}$ to report effect size.

RESULTS

Table 2 shows the mean accuracy and voice-onset times (correct responses only) for the 2 tasks by group. The patients performed worse on Task 1 (visuospatial grid navigation) than on Task 2 (sequential number subtraction). The controls did not perform at ceiling and showed similar accuracy on both tasks.

Analyses of Sensitivity (d')

Figure 2 shows mean d' values by task and group.

A 2 (Task: Visuospatial grid navigation, Sequential number subtraction) × 2 (Group: PD, control) × 3 (Sequence length: 2, 3, 4) mixed analysis of variance (ANOVA) showed significant main effects of Task: $F_{1,62} = 7.47$, $P = 0.008$, $\eta p^2 = 0.11$, and of Group: $F_{1,62} = 20.89$,

TABLE 2. Accuracy and Voice-Onset Time for the Parkinson Disease (PD) and Control Groups Across Tasks

	Mean Accuracy (% Correct) (SD)			Mean Voice-Onset Time (msec) (SD)		
	Sequence Length			Sequence Length		
	2	3	4	2	3	4
Task 1: Visuospatial grid navigation						
PD	85.58 (8.98)	89.9 (9.03)	91.35 (8.28)	1113.62 (196.29)	1101.5 (292.13)	1196.79 (254.62)
Control	96.25 (3.14)	98.13 (3.57)	95.94 (3.67)	1055.12 (166.96)	1009.21 (157.19)	959.71 (139.67)
Task 2: Sequential number subtraction						
PD	96.16 (6.1)	95.19 (2.74)	97.12 (3.24)	1108.37 (164.22)	1072.92 (381.47)	1035.62 (203.76)
Control	97.19 (4.74)	96.25 (4.25)	95.94 (4.66)	1005.87 (172.3)	951.87 (165.87)	944.55 (208.77)

msec = milliseconds. **SD** = standard deviation.

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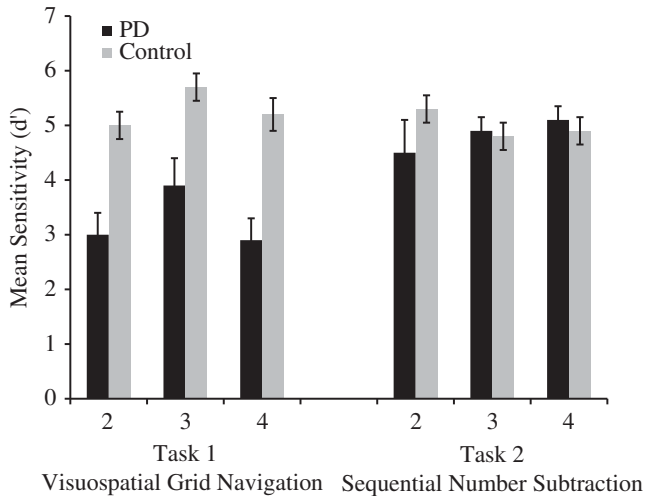

FIGURE 2. Mean sensitivity (*d'*) as a function of task and sequence length. Error bars show the standard error of the mean.

$P < 0.001$, $\eta p^2 = 0.11$. There was also a significant Task × Group interaction: $F_{1,62} = 15.05$, $P < 0.001$, $\eta p^2 = 0.20$, and a significant Sequence length × Task interaction: $F_{2,124} = 3.290$, $P = 0.041$, $\eta p^2 = 0.05$.

To explore the interactions further, we used separate 2 (Task) × 3 (Sequence length) ANOVAs for each group. The control group showed no significant main effects or interactions. In contrast, the PD group showed a significant main effect of Task: $F_{1,24} = 13.68$, $P < 0.001$, $\eta p^2 = 0.36$. This showed that unlike the controls, the patients were significantly worse at visuospatial grid navigation than at sequential number subtraction.

We performed additional analyses separately for each task:

Task 1: Visuospatial Grid Navigation

A 2 (Group: PD, control) × 3 (Sequence length: 2, 3, 4) mixed ANOVA showed a significant main effect of Group: $F_{1,31} = 42.223$, $P < 0.001$, $\eta p^2 = 0.58$, confirming that the patients were impaired relative to the controls. There was also a significant main effect of Sequence length: $F_{2,62} = 4.74$, $P = 0.012$, $\eta p^2 = 0.13$, but no Group × Sequence length interaction: $F_{2,62} = 0.58$, $P = 0.57$, $\eta p^2 = 0.018$.

Additional planned comparisons showed that the patients performed worse than the controls at all 3 sequence lengths: Sequence length 2: $F_{1,32} = 21.58$, $P < 0.001$, $\eta p^2 = 0.41$; Sequence length 3: $F_{1,32} = 14.32$, $P < 0.001$, $\eta p^2 = 0.32$; and Sequence length 4: $F_{1,32} = 25.96$, $P < 0.001$, $\eta p^2 = 0.46$.

Task 2: Sequential Number Subtraction

A 2 (Group: PD, control) × 3 (Sequence length: 2, 3, 4) mixed ANOVA found no significant effects of Group: $F_{1,31} = 0.21$, $P = 0.65$, $\eta p^2 = 0.007$, or Sequence length: $F_{2,62} = 0.343$, $P = 0.711$, $\eta p^2 = 0.011$. This shows that the patients performed as well as the controls in sequential number subtraction.

Planned Analyses of Voice-Onset Times

We analyzed the mean voice-onset times for correct responses after removing outliers (6.16% of trials). A 2 (Group: PD, control) × 2 (Task: 1, 2) × 3 (Sequence length: 2, 3, 4) mixed ANOVA showed significant main effects of Group: $F_{1,62} = 9.02$, $P = 0.004$, $\eta p^2 = 0.24$, and Sequence length: $F_{2,124} = 23.74$, $P < 0.001$, $\eta p^2 = 0.28$, but no effect of Task: $F_{1,62} = 3.31$, $P = 0.074$, $\eta p^2 = 0.05$. There were significant 2-way interactions for Task × Sequence length: $F_{2,124} = 5.70$, $P = 0.004$, $\eta p^2 = 0.084$, and for Group × Sequence length: $F_{2,124} = 4.23$, $P = 0.02$, $\eta p^2 = 0.06$. The 3-way Group × Task × Sequence length interaction was also significant: $F_{2,124} = 4.86$, $P = 0.009$, $\eta p^2 = 0.17$.

We investigated the interactions further by running separate ANOVAs for each task:

Task 1: Visuospatial Grid Navigation

A 2 (Group) × 3 (Sequence length) mixed ANOVA showed that the patients were significantly slower than the controls: $F_{1,31} = 7.83$, $P = 0.009$, $\eta p^2 = 0.202$. There was a significant main effect of Sequence length: $F_{2,62} = 32.21$, $P > 0.001$, $\eta p^2 = 0.51$, and a Group × Sequence length interaction: $F_{2,62} = 10.944$, $P > 0.001$, $\eta p^2 = 0.261$. The patients were significantly slower than the controls on Sequence length 2: $F_{1,31} = 14.18$, $P = 0.001$, $\eta p^2 = 0.31$, but not on Sequence length 3: $F_{1,31} = 2.86$, $P = 0.10$, $\eta p^2 = 0.08$, or Sequence length 4: $F_{1,31} = 3.99$, $P = 0.055$, $\eta p^2 = 0.11$.

Task 2: Sequential Number Subtraction

A 2 (Group) × 3 (Sequence length) ANOVA showed no significant main effect of Group: $F_{1,31} = 0.065$, $P = 0.80$, $\eta p^2 = 0.002$, or of Sequence length: $F_{2,62} = 0.646$, $P = 0.528$, $\eta p^2 = 0.02$.

Further Analyses

For the PD group, we also conducted bivariate correlations between the baseline measures and the 2 experimental tasks (Table 3).

We found that disease duration and severity (Hoehn and Yahr score) had significant correlations with Wechsler Abbreviated Scale of Intelligence Full IQ and with both the verbal and performance subtests. The patients' performance on Task 1 (Visuospatial grid navigation) was positively correlated with their spatial short-term memory but not with the Benton Visual Retention Test. Task 2 (Sequential number subtraction) was negatively correlated with Benton Visual Retention Test scores. There was no significant correlation in performance between the 2 experimental tasks.

Separate analyses of the spatial short-term memory baseline task showed that response accuracy was high for both the PD (mean = 92.48; SD = 2.38) and control groups (mean = 91.74, SD = 4.68). The difference in means was not significant: $U = 114.5$, $P = 0.57$, $r = 0.11$. This shows that the patients were able to encode and maintain memory for a single spatial location over 5 seconds, a duration that exceeded the maximum interstimulus interval on the 2 tasks. Thus, although spatial

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TABLE 3. Bivariate Correlations for Screening Measures and Experimental Tasks

	Hoehn and Yahr Stage[1]	Disease Duration	Age	Spatial Short-Term Memory	Task 1: Visuospatial Grid Navigation	Task 2: Sequential Number Subtraction	MMSE[2]	BVRT[3]	WASI[4] Verbal	WASI[4] Performance
Disease duration	0.678*									
Age	−0.201	0.158								
Spatial short-term memory	0.256	0.427	0.405							
Task 1: Visuospatial grid navigation	0.384	0.168	0.159	0.681*						
Task 2: Sequential number subtraction	0.467	0.496	0.172	0.073	0.016					
Mini-Mental State Examination (MMSE)[2]	0.21	0.283	−0.524	0.121	0.237	−0.101				
Benton Visual Retention Test (BVRT)[3]	−0.251	−0.42	−0.176	−0.141	0.023	−0.716**	0.09			
Wechsler Abbreviated Scale of Intelligence (WASI)[4]										
Verbal	0.306	0.403	−0.015	−0.058	0.203	−0.015	0.022	−0.276		
Performance	−0.13	−0.038	−0.359	−0.512	−0.137	−0.231	0.508	0.134	0.278	
Full	0.065	0.321	−0.17	−0.131	0.034	−0.261	0.341	−0.107	0.765**	0.706**

*Correlation is significant at the 0.05 level (2-tailed).
**Correlation is significant at the 0.01 level (2-tailed).
[1]Hoehn and Yahr, 1967. [2]Folstein et al, 1975. [3]Benton Sivan, 1992. [4]Wechsler, 1999.

memory correlated with performance in Task 1, impairment to spatial memory cannot account for the patients' poorer performance than the controls' on visuospatial grid navigation.

DISCUSSION

Our results show a clear dissociation in the performance of patients with PD between visuospatial grid navigation and sequential number subtraction. The patients performed worse than the controls on the visuospatial grid navigation task, but as well as the controls in sequential number subtraction across identical trial sequence lengths. The controls performed equally well on both tasks, but below ceiling, suggesting that the 2 tasks were well matched in difficulty.

These findings provide new evidence about the functional substrates of visuospatial processing deficits in PD. In particular, our results are inconsistent with the hypothesis that visuospatial impairment in PD arises from a more general, nonvisuospatial deficit in sequencing or serial chaining. If this hypothesis were true, we would expect patients who performed poorly on visuospatial tasks like grid navigation (or mental rotation) also to exhibit problems in nonvisuospatial sequencing tasks of equivalent difficulty, such as sequential number subtraction. This is not what we found.

Our data suggest that patients with PD can have deficits in visuospatial processing without having a generalized impairment in sequence processing. It is important to note, however, that our evidence of a dissociation between visuospatial processing and nonvisuospatial sequencing does not preclude the possibility of separate impairments to sequencing or serial chaining in PD, or their co-occurrence with visuospatial task impairments in some patients.

More broadly, our results support other evidence of visuospatial impairment in PD (Amick et al, 2006; Boller et al, 1984; Cronin-Golomb and Amick, 2001; Duncombe et al, 1994; Kerai et al, 2012; Lee et al, 1998; Levin et al, 1991; Montse et al, 2001; Pillon et al, 1991; Raskin et al, 1990; Sawamoto et al, 2002), and invite further speculation about the underlying functional deficit responsible for these impairments.

Kerai et al (2012) concluded that, within the context of mental rotation tasks, patients with PD show evidence of impairment to cognitive processes involved in the computation of spatial transformations—what we have defined as domain-general routines that compute and execute mappings between visual images or spatial locations (Leek and Johnston, 2009). Kerai and colleagues' conclusion was supported by their observation of slower reaction times in patients with PD than controls; the greater the stimulus angular disparity, the slower the patients' reaction times. This pattern is consistent with a slower rate of image normalization, as might be expected from an underlying deficit to mechanisms supporting spatial transformation.

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Our current finding of a deficit in visuospatial grid navigation is also consistent with this account, in that successful task performance requires the computation of a spatial transformation after each arrow cue, ie, to determine the direction of movement from the old to a new grid location.

In terms of the supporting neurophysiology, there is evidence that visuospatial processing is supported by a network of brain regions that are known to include prefrontal areas such as the dorsolateral prefrontal cortex, supplementary motor area, and superior parietal cortex/anterior intraparietal area (eg, Johnston et al, 2004; Richter et al, 2000; Vingerhoets et al, 2001; Windischberger et al, 2003). This same network is also associated with the basal ganglia, which share connections with the dorsolateral prefrontal cortex, supplementary motor area, and parietal cortex, providing a link between the underlying pathology of PD and visuospatial processing (eg, Clower et al, 2005; Lehéricy et al, 2004). Thus, visuospatial deficits in PD are likely to arise from dysfunction within this corticostriatal dopaminergic network.

In summary, patients with PD and age-matched neurologically normal controls performed a visuospatial grid navigation task requiring sequential spatial transformations, and a control task of serial number subtraction designed to assess nonvisuospatial sequencing. The tasks were alike in structure and difficulty. The patients were impaired on the visuospatial task, but not in sequential number subtraction. This finding suggests that visuospatial processing impairments in PD do not derive from a general impairment affecting the sequencing or serial chaining of mental operations. We argue that visuospatial deficits in PD result from impairments to spatial transformation routines involved in the computation of mappings between spatial locations. These routines are mediated by dopaminergic pathways linking the basal ganglia, prefrontal cortex, supplementary motor area, and parietal cortex.

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